

PERUGIA, Italy

WMO: 161810

Lat: 43.0925N

Lon: 12.5047E

Elev: 204

StdP: 98.90

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	-2.1	-8.0	2.0	1.3	-6.2	2.3	0.8	11.4	5.5	10.1	6.1	2.2	0	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	35.2	20.0	34.0	19.8	32.8	19.7	22.0	30.3	21.3	29.7	20.7	29.2	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.7	24.3	18.9	14.1	24.1	18.0	13.3	23.4	65.1	30.5	62.8	30.0	60.5	29.4	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.3	7.3	-6.4	38.2	1.3	1.1	-7.3	39.0	-8.0	39.7	-8.7	40.3	-9.7	41.1		
(5)			-6.7	23.2	1.3	1.0	-7.6	24.0	-8.4	24.5	-9.1	25.1	-10.1	25.8		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	5.3	6.3	9.2	12.4	16.3	21.3	24.3	24.3	20.0	15.5	10.1	6.2
(7)		DBStd	7.39	2.89	3.03	3.04	2.85	2.80	3.16	2.41	2.46	2.98	3.02	3.71	3.71
(8)		HDD10.0	491	147	106	52	11	1	0	0	0	3	44	127	
(9)		HDD18.3	2022	404	336	284	178	76	9	0	1	17	94	248	376
(10)		CDD10.0	2067	2	4	27	83	196	339	444	444	301	173	47	9
(11)		CDD18.3	557	0	0	0	1	13	98	186	186	68	6	0	0
(12)		CDH23.3	6535	0	0	0	21	202	1206	2309	2118	634	46	0	0
(13)		CDH26.7	2974	0	0	0	2	40	516	1164	1059	188	5	0	0
(14)	Wind	WSAvg	2.9	2.7	3.2	3.4	3.0	2.8	2.8	3.1	2.8	3.0	2.8	2.6	2.7
(15)	Precipitation	PrecAvg	812	52	58	62	73	62	56	37	48	89	96	104	75
(16)		PrecMax	1151	150	136	135	120	150	122	113	135	195	221	210	171
(17)		PrecMin	478	9	4	5	18	28	14	0	0	37	8	7	2
(18)		PrecStd	150	33	35	34	25	30	26	23	31	33	47	59	45
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	14.4	17.1	21.8	25.9	30.2	35.8	37.2	37.2	33.1	27.0	20.8	16.2	
(20)		0.4% MCWB	10.8	10.6	12.0	15.9	18.3	20.3	19.7	19.8	19.1	17.4	16.0	13.5	
(21)		2% DB	13.0	14.9	19.8	23.1	27.7	33.8	35.8	35.8	31.0	24.8	18.6	14.8	
(22)		2% MCWB	9.9	9.8	11.8	14.4	17.5	19.6	19.5	20.1	18.7	16.9	14.9	12.7	
(23)		5% DB	11.8	13.1	17.2	21.1	25.8	32.0	34.1	33.9	28.9	22.9	17.0	13.2	
(24)		5% MCWB	9.1	9.3	11.1	13.8	16.9	19.8	19.7	20.1	18.4	16.6	14.0	11.3	
(25)		10% DB	10.2	11.9	15.4	19.1	23.9	30.0	32.8	32.1	27.0	21.1	15.2	12.0	
(26)		10% MCWB	8.3	8.6	10.4	12.7	16.1	19.2	19.5	19.9	17.7	16.2	12.8	10.2	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	11.7	12.2	13.8	17.0	19.7	22.2	22.7	23.2	21.4	19.1	17.3	14.1	
(28)		0.4% MCDB	13.8	15.1	18.7	23.3	27.7	30.9	30.8	30.7	27.8	22.6	19.5	15.8	
(29)		2% WB	10.4	10.9	12.7	15.4	18.5	21.2	21.7	22.2	20.2	18.0	15.5	12.7	
(30)		2% MCDB	12.2	13.4	17.3	21.0	25.6	30.5	30.2	30.4	27.0	22.0	17.8	14.4	
(31)		5% WB	9.7	10.0	11.7	14.3	17.5	20.4	21.2	21.4	19.5	17.3	14.2	11.7	
(32)		5% MCDB	11.1	12.2	16.1	19.6	24.2	29.7	29.8	29.6	26.1	21.4	16.3	13.3	
(33)		10% WB	8.6	9.0	10.9	13.4	16.6	19.6	20.5	20.8	18.8	16.6	13.2	10.4	
(34)		10% MCDB	10.0	11.2	14.8	18.1	22.3	28.2	29.3	29.0	24.9	20.5	15.3	11.8	
(35)	Mean Daily Temperature Range	MDBR	7.8	9.6	11.0	11.7	12.3	14.0	15.1	14.5	12.4	10.3	8.3	7.2	
(36)		5% DB MCDBR	9.5	12.0	15.0	15.8	15.9	17.2	17.5	17.1	15.4	12.6	9.5	7.5	
(37)		5% WB MCWBR	6.8	7.7	8.4	8.4	7.5	5.9	5.3	5.3	5.5	6.4	6.1	5.7	
(38)		5% WB MCDBR	7.6	9.9	12.6	13.3	14.0	15.7	15.0	14.4	12.4	10.3	8.4	6.6	
(39)	Clear-Sky Solar Irradiance	taub	0.334	0.356	0.391	0.428	0.432	0.430	0.418	0.416	0.404	0.387	0.361	0.333	
(40)		taud	2.429	2.342	2.253	2.165	2.197	2.249	2.284	2.307	2.326	2.359	2.371	2.443	
(41)		Ebn at Noon	794	831	843	838	844	845	851	841	821	786	747	758	
(42)		Edh at Noon	77	98	121	143	142	135	129	122	112	96	81	70	
(43)	All-Sky Solar Radiation	RadAvg	1.54	2.36	3.51	4.67	5.78	6.68	6.97	6.05	4.33	2.86	1.72	1.37	
(44)		RadStd	0.19	0.29	0.48	0.45	0.56	0.34	0.36	0.41	0.26	0.28	0.18	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82	N/A

Nomenclature: See separate page